

Vital Sign Description: Blood Pressure

Blood pressure is a vital sign that measures the force of blood on the walls of arteries as it circulates the body. It is an important indicator of how well the heart and cardiovascular system are functioning. Proper blood flow is essential because it delivers oxygen and nutrients to the body's organs and tissues.

Blood pressure is recorded as two numbers: systolic blood pressure (top number), which measures the pressure your blood is pushing against your artery walls when the heart beats, and diastolic blood pressure (bottom number), which measures the pressure your blood is pushing against your artery walls while the heart muscle rests between beats (1). **A normal blood pressure reading in most adults ranges from 90/60 mmHg to 120/80 mmHG (2).** However, the normal range can vary depending on the age of the patient. Blood pressure is not constant and can change depending on recent activity level, emotional state, age, body position, and medications. Readings above the normal range may indicate hypertension, a major risk factor for cardiovascular disease, while readings below the normal range may suggest hypotension, which can reduce blood flow to vital organs leading to dizziness, fainting and shock (3). See figure 1 for Blood Pressure Categories.

In triage, blood pressure helps to determine how stable or critical a patient's condition may be. It allows healthcare professionals to quickly identify and prioritize patients who may be at high risk of deterioration and require urgent medical attention. When combined with a patient's symptoms, abnormal blood pressure readings can elevate their triage priority to prevent life threatening complications. Overall, blood pressure monitoring can support immediate treatment decisions and help assess whether a patient's condition is improving or worsening over time.

Figure 1: Blood Pressure Categories chart

BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (top/upper number)	and/or	DIASTOLIC mm Hg (bottom/lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
STAGE 1 HYPERTENSION (High Blood Pressure)	130 – 139	or	80 – 89
STAGE 2 HYPERTENSION (High Blood Pressure)	140 OR HIGHER	or	90 OR HIGHER
SEVERE HYPERTENSION (If you don't have symptoms*, call your health care professional.)	HIGHER THAN 180	and/or	HIGHER THAN 120
HYPERTENSIVE EMERGENCY (If you have any of these symptoms*, call 911.)	HIGHER THAN 180	and/or	HIGHER THAN 120

*symptoms: chest pain, shortness of breath, back pain, numbness, weakness, change in vision or difficulty speaking

References

1. American Heart Association. Understanding blood pressure readings [Internet]. Available from: <https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings>
2. Cleveland Clinic. *Low blood pressure (hypotension)* [Internet]. Available from: <https://my.clevelandclinic.org/health/diseases/21156-low-blood-pressure-hypotension>
3. Cleveland Clinic. Blood pressure: diagnosis and overview [Internet]. Available from: <https://my.clevelandclinic.org/health/diagnostics/17649-blood-pressure>
4. World Health Organization. *Hypertension fact sheet* [Internet]. Available from: <https://www.who.int/news-room/fact-sheets/detail/hypertension>